

NORTHEAST IOWA, IA, USA

WMO: 725463

Lat: 43.073N

Lon: 92.611W

Elev: 343

StdP: 97.27

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 14966

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.6	-20.1	-27.2	0.3	-21.7	-24.1	0.4	-19.6	14.3	-10.3	12.6	-8.5	4.0	310	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	32.3	24.1	30.3	23.2	28.8	22.6	26.2	29.6	25.0	28.6	23.9	27.5	5.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	21.1	28.4	23.8	19.5	27.1	22.6	18.1	26.3	83.4	29.2	78.1	28.4	73.6	27.2	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.6	9.1	-28.1	34.4	3.5	1.7	-30.6	35.6	-32.7	36.6	-34.6	37.5	-37.1	38.7		
(5)			-27.5	27.8	3.5	1.1	-30.1	28.6	-32.2	29.2	-34.2	29.8	-36.7	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-7.4	-5.3	1.0	8.8	15.3	20.8	22.3	21.0	17.4	10.2	2.0	-4.7		
	DBStd	11.87	7.13	6.84	6.89	5.63	4.83	3.48	2.98	2.78	4.40	5.26	5.79	6.47		
	HDD10.0	2124	539	430	292	87	9	0	0	0	3	63	246	455		
	HDD18.3	3964	798	663	539	288	119	16	4	7	70	258	489	713		
	CDD10.0	1583	0	0	13	51	174	323	382	341	223	68	7	0		
	CDD18.3	381	0	0	2	2	25	88	128	90	41	5	0	0		
	CDH23.3	3255	0	0	11	45	262	769	1094	666	351	56	1	0		
	CDH26.7	947	0	0	2	11	81	246	340	159	96	12	0	0		
(14) Wind	WSAvg	4.3	4.9	4.9	5.0	5.3	4.9	3.9	3.2	2.8	3.4	4.3	4.7	4.7		
(15) Precipitation	PrecAvg	868	22	23	52	85	110	124	117	105	93	60	45	27		
	PrecMax	1306	50	72	137	196	259	290	469	345	318	147	136	81		
	PrecMin	421	2	0	6	17	40	35	20	19	15	6	0	5		
	PrecStd	176	13	16	29	38	48	63	75	75	69	33	34	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	12.8	24.1	27.8	32.5	33.7	33.7	32.4	32.3	28.7	20.9	12.9		
		MCWB	5.8	10.2	16.7	17.1	20.7	23.9	26.0	24.2	22.7	18.4	13.9	10.7		
	2%	DB	5.2	8.7	18.9	23.8	28.7	31.6	32.2	30.0	29.1	24.8	17.0	7.8		
		MCWB	3.1	7.1	14.0	14.8	18.8	22.8	25.8	24.8	21.8	16.9	12.2	6.1		
	5%	DB	2.8	5.2	14.8	21.0	26.3	29.7	30.1	28.0	27.3	22.1	12.9	5.2		
		MCWB	1.7	3.4	10.7	13.1	17.8	22.0	24.6	23.3	20.4	16.0	9.1	3.5		
	10%	DB	1.4	2.9	10.0	17.9	23.7	27.6	28.6	27.2	25.1	18.8	10.2	2.7		
		MCWB	0.6	1.9	7.5	11.6	16.9	20.6	23.8	22.3	19.1	14.4	7.6	1.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	10.5	17.2	18.5	23.6	26.6	28.5	27.2	25.4	21.8	15.5	11.8		
		MCDB	9.3	12.5	23.6	26.0	28.4	30.8	32.0	30.0	29.2	25.9	18.7	13.7		
	2%	WB	3.6	7.1	14.7	16.3	21.3	24.8	26.7	25.5	23.4	19.2	12.8	6.6		
		MCDB	5.2	8.8	18.1	22.0	26.5	29.1	29.7	28.8	27.1	22.1	15.9	7.7		
	5%	WB	2.0	3.8	10.8	14.6	19.6	23.4	25.4	24.3	22.1	16.9	10.1	3.8		
		MCDB	2.9	5.2	14.6	19.4	24.3	27.8	28.9	27.3	26.0	19.8	12.9	5.3		
	10%	WB	0.7	2.2	7.3	12.6	18.1	22.1	24.1	23.1	20.6	14.8	7.7	2.0		
		MCDB	1.5	3.3	9.7	17.1	21.8	26.4	27.8	25.9	23.5	18.7	9.9	2.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.3	8.5	10.9	10.9	10.5	10.3	10.4	11.9	11.0	8.9	7.5		
		MCDBR	9.6	9.4	13.6	16.2	14.4	13.0	11.5	11.3	13.8	14.7	13.3	9.4		
	5% WB	MCWBR	8.2	7.7	8.7	8.8	6.9	6.3	6.6	7.1	7.2	8.1	8.7	7.7		
		MCWBR	9.1	9.0	12.9	13.6	11.8	11.1	10.2	9.9	11.4	11.8	12.2	8.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.274	0.299	0.321	0.369	0.397	0.411	0.429	0.410	0.377	0.333	0.294	0.271		
		taud	2.390	2.321	2.392	2.299	2.286	2.287	2.225	2.288	2.346	2.460	2.495	2.428		
	Ebn at Noon	Ebn at Noon	869	898	922	896	877	862	841	846	849	852	843	841		
		Edh at Noon	77	98	104	124	130	130	137	125	109	86	70	68		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.81	2.85	3.78	4.63	5.49	6.09	6.26	5.36	4.35	2.83	1.90	1.41		
	RadStd	RadStd	0.19	0.27	0.39	0.44	0.40	0.48	0.32	0.36	0.31	0.38	0.16	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page